

Mathematical background

Branching processes and generating-function methods

Adam Aldridge

Abstract

This chapter develops the generating-function methods used later to analyse small replicating populations and intracellular pathogen dynamics. After a short common vocabulary of Markov chains, probability generating functions, branching criticality, and forward (master) equations, a binary Galton–Watson process provides an exact account of extinction, establishment from a finite cohort, critical survival and total progeny, and the subcritical limiting law conditioned on survival. Its late-generation survival amplitude is identified with the classical Kolmogorov constant, represented both by a convergent product and by a Koenigs linearisation. The discrete model is then compared with the continuous-time linear birth–death process, for which the corresponding conditional limit is explicit. Finally, the same probabilistic structure is extended to bivariate absorption, death, and replication models. Their forward equations lead to first-order partial differential equations for multivariate probability generating functions; independence and characteristics yield exact, normalised solutions and reliable coefficient formulae.

Contents

1	Scope and modelling perspective	3
2	Mathematical preliminaries	4
2.1	Discrete-time Markov chains	4
2.2	Continuous-time Markov chains	4
2.3	Probability generating functions	5
2.4	Criticality for branching	6
2.5	Forward equations and the master equation	6
3	Galton–Watson survival and the limiting conditional law	7
3.1	Offspring PGFs and functional iteration	7
3.2	The binary death-or-division process	8
3.3	Critical survival and extinction time	8
3.4	Critical total progeny	9
3.5	Subcritical survival and the Kolmogorov constant	10
3.6	Yaglom limit and conditional moments	11
3.7	Koenigs linearisation and certified computation	12
4	Initial cohorts, catastrophe exposure, and continuous-time comparison	13
4.1	Survival from a finite cohort	13
4.2	A state-dependent catastrophe is not a mean-field hazard	14

4.3	The continuous-time linear birth–death process	15
4.4	Why the discrete and continuous amplitudes differ	16
5	Multivariate PGFs and the method of characteristics	16
5.1	A two-compartment Markov model	16
5.2	Absorption without intracellular dynamics	17
5.3	Absorption followed by intracellular death	18
5.4	Absorption followed by intracellular birth and death	18
5.5	Coefficient extraction and moment checks	20
5.6	Synthesis	20

1 Scope and modelling perspective

Deterministic rate equations describe a population well once its numbers are large, and badly once they are not. When counts are of order one—in early infection, founding inocula, and small-population regimes more generally—extinction is an event of genuine probability rather than a limiting curiosity, and the typical size of those lineages that are still alive can differ sharply from the unconditional mean. A population may have a positive expected growth rate and nevertheless disappear because its first few lineages fail; conversely, a subcritical population is certain to become extinct but may approach a stable distribution when observed conditional on still being present. The same distinction reappears when pathogens move between extracellular and intracellular compartments. Mean equations describe average flux, whereas extinction, absorption, and the full distribution of intracellular burden require a stochastic account.

Branching processes supply the natural language for that setting, because they track whole lineages rather than an averaged density. The classical discrete-time construction is the Galton–Watson process of independent generational replacement; continuous-time analogues replace generations by exponential clocks and linear rates. Both sit inside a broader Markovian framework in which absorption encodes extinction, rupture, or removal from a compartment, and in which conditioning on non-absorption produces the quasi-stationary pictures used later in the thesis.

Two classical regimes will recur throughout. In slightly supercritical branching, the chance of eventual survival is highly sensitive to the initial number of individuals—a fact with a direct reading for early infection. In the subcritical regime extinction is certain, and yet the population size conditional on non-extinction can settle into a quasi-stationary law; the reciprocal of that conditional mean is explicit for continuous-time linear birth–death, whereas in discrete time it is organised by the Kolmogorov constant $A(p)$.

Probability generating functions (PGFs) provide a common analytic language for these questions. In discrete time, functional iteration of an offspring PGF records the descendants of a Galton–Watson process. In continuous time, a multivariate PGF converts the forward equations of a Markov chain into a first-order partial differential equation (PDE). Both constructions rely on the same branching property: descendants of distinct initial individuals evolve independently, so the PGF for an initial cohort of size k is the k th power of the one-ancestor PGF.

The chapter develops this framework in three stages, after the short common vocabulary of [section 2](#). First, [section 3](#) treats a binary death-or-division process and separates finite-time survival, ultimate establishment, and the limiting conditional law. The late-generation survival amplitude is the Kolmogorov constant; its exact product and Koenigs representations give complementary probabilistic and dynamical interpretations. Second, [section 4](#) quantifies the effects of an initial cohort and a state-dependent catastrophe, compares discrete and continuous time, and identifies why their conditional means need not agree away from criticality. Third, [section 5](#) derives exact bivariate PGFs for absorption, intracellular death, and intracellular replication. The resulting model hierarchy is deliberately ladderred: each additional mechanism is introduced only after the simpler process and its assumptions are explicit.

What follows is written so that a reader with standard undergraduate probability and ordinary differential equations can proceed without the rest of the thesis in hand. Forward references to later chapters are kept minimal; the aim is a reusable stochastic structure rather than a catalogue of which later section cites which formula. Throughout, Z_n denotes a discrete-generation population, whereas X_t and Y_t denote intracellular and extracellular

counts in continuous time. The symbol p is reserved for the probability of division in the binary Galton–Watson process; rates are denoted by Greek letters. Extinction is an absorbing state, and all stochastic lineages are independent unless a global catastrophe mechanism is stated explicitly.

2 Mathematical preliminaries

This section fixes a short common vocabulary. The aim is not a course in stochastic processes but a set of names and pictures that the rest of the chapter—and the continuous-time models later in the thesis—can use without redefinition. We concentrate on four things: discrete- and continuous-time Markov structure, generating functions, the elementary trichotomy of branching criticality, and the forward (master) equations that will later become PGF partial differential equations.

2.1 Discrete-time Markov chains

A discrete-time Markov chain on a countable state space $\mathcal{S}$ is a sequence of random variables $(X_n)_{n \geq 0}$ taking values in $\mathcal{S}$ and satisfying the Markov property: the conditional law of the next state depends on the past only through the present. Writing $p(i, j)$ for the one-step transition probability from i to j , that property reads

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) = p(i, j), \quad (2.1)$$

whenever the conditioning event has positive probability.

A state $a \in \mathcal{S}$ is *absorbing* if $p(a, a) = 1$; once the chain enters such a state it remains there forever. Extinction of a population, and later the rupture or catastrophe events of birth–death–catastrophe models, will all be encoded by absorption of this kind, sometimes after enlarging the state space with a cemetery state.

The Galton–Watson process treated in [section 3](#) is a discrete-time Markov chain of exactly this type: it lives on $\mathcal{S} = \{0, 1, 2, \dots\}$ and is absorbed at 0. Its one-step transitions are determined by an offspring distribution rather than by an arbitrary matrix $p(i, j)$, but the Markov and absorption language carries over unchanged.

2.2 Continuous-time Markov chains

A continuous-time Markov chain (CTMC) on a countable state space replaces discrete steps by holding times and jumps. While the process occupies state i it waits an exponential time whose rate q_i may depend on i , and then jumps to a different state j with probabilities determined by the off-diagonal rates $q(i, j)$. The infinitesimal description, for small $\Delta t > 0$, is

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q(i, j) \Delta t + o(\Delta t), \quad j \neq i, \quad (2.2)$$

$$\mathbb{P}(X_{t+\Delta t} = i \mid X_t = i) = 1 - q_i \Delta t + o(\Delta t), \quad (2.3)$$

with

$$q_i = \sum_{j \neq i} q(i, j),$$

assumed finite. The Markov property again holds, now with respect to continuous time.

In population models these rates are typically linear, or at least affine, in the current count. A single-type birth–death process on $\{0, 1, 2, \dots\}$ with per capita birth rate β and death rate μ therefore has, when $X_t = n \geq 1$,

$$q(n, n+1) = \beta n, \quad q(n, n-1) = \mu n, \quad q_n = (\beta + \mu)n, \quad (2.4)$$

with 0 absorbing if there is no immigration. A catastrophe or killing mechanism adds a further transition out of every $n \geq 1$, at a rate proportional to n or to some more general function of n ; that structure is the continuous-time backbone of the birth–death–catastrophe analysis later in the thesis. The two-compartment absorption models of [section 5](#) are CTMCs of the same type, with rates linear in the extracellular and intracellular counts separately.

Two elementary facts will be used repeatedly. First, the holding time in i is memoryless, so the future after a jump depends only on the arrival state. Second, absorption is again well defined: if $q_a = 0$, then a is absorbing in continuous time. Pathwise, a CTMC may be constructed from its jump chain—the discrete-time sequence of visited states—together with independent exponential clocks. We shall seldom need that construction explicitly, but it is worth having in view, since it explains why discrete-time branching intuition and continuous-time rate models sit in the same family.

2.3 Probability generating functions

For a random variable N taking values in $\{0, 1, 2, \dots\}$, the probability generating function is

$$G(z) := \mathbb{E}[z^N] = \sum_{k=0}^{\infty} \mathbb{P}(N = k) z^k, \quad |z| \leq 1. \quad (2.5)$$

When the series converges in a neighbourhood of the origin, derivatives at 0 recover the point masses, and

$$\mathbb{E}[N] = G'(1), \quad \mathbb{E}[N(N-1)] = G''(1), \quad (2.6)$$

whenever those moments exist, with one-sided derivatives at $z = 1$ if necessary.

The reason generating functions earn their place here is that they turn independent sums into products, and thereby turn many branching calculations into functional iteration or ordinary differential equations. If each individual in a Galton–Watson generation produces offspring independently with generating function $f(z)$, then the population size after one generation from a single ancestor has generating function f , and after n generations the n -fold compositional iterate $f^{\circ n}$. In continuous time the PGF of a CTMC's state,

$$G(z, t) = \mathbb{E}[z^{X_t}],$$

often satisfies a first-order partial differential equation obtained from the master equation below. That second route is opened for linear birth–death in [section 4.3](#) and developed fully by the method of characteristics in [section 5](#).

For a pair of non-negative integer-valued random variables (X, Y) , the bivariate extension

$$G(x, y) = \mathbb{E}[x^X y^Y] = \sum_{i,j} \mathbb{P}(X = i, Y = j) x^i y^j \quad (2.7)$$

will be the working object of the compartmental models. Marginal moments again arise by differentiation at $(1, 1)$, and independence of lineages continues to produce powers: if N independent copies of a one-particle process are superposed, the joint PGF is the one-particle PGF raised to the N th power.

2.4 Criticality for branching

Branching mechanisms are organised by a single comparison of mean growth to replacement. The same trichotomy appears in discrete and continuous time; only the bookkeeping of the mean changes.

Discrete time. Suppose each individual has offspring mean $m = \mathbb{E}[\xi]$, where ξ is a generic offspring count, and exclude the trivial law concentrated at 1. The associated Galton–Watson process is

- *subcritical* if $m < 1$,
- *critical* if $m = 1$,
- *supercritical* if $m > 1$.

In the binary death-or-division case used throughout much of this chapter, an individual is replaced by two offspring with probability p and by zero with probability $1 - p$, so that $m = 2p$. Subcriticality is then $p < \frac{1}{2}$, criticality $p = \frac{1}{2}$, and supercriticality $p > \frac{1}{2}$.

Continuous time. For a linear birth–death process with per capita rates β and μ , the mean $M(t) = \mathbb{E}[X_t]$ obeys

$$\frac{dM}{dt} = (\beta - \mu)M \quad (2.8)$$

when it exists, so the net growth rate $\beta - \mu$ plays the role of $\log m$ per unit time. The process is called subcritical, critical, or supercritical according as $\beta - \mu$ is negative, zero, or positive, again excluding degenerate cases with no randomness.

Why the trichotomy governs what follows. Extinction probabilities, the existence of a non-trivial survival event, and the nature of population size conditional on non-extinction all change character across the critical threshold. In the subcritical regime extinction is certain, yet the conditional distribution of size given survival can settle into a quasi-stationary picture; near criticality those conditional sizes become large; above criticality there is a positive probability of indefinite growth. The sections on discrete and continuous branching specialise these regimes in turn, and the constant $A(p)$ of the discrete theory will live naturally in the subcritical half-line $p \in [0, \frac{1}{2})$.

2.5 Forward equations and the master equation

Let $p_n(t) = \mathbb{P}(X_t = n)$ for a CTMC on $\{0, 1, 2, \dots\}$. Balancing probability flux into and out of each state yields the forward Kolmogorov equations, often called the *master equation* in the physical literature. For the linear birth–death process with rates β, μ as in [equation \(2.4\)](#) they read

$$\frac{dp_0}{dt} = \mu p_1, \quad (2.9)$$

$$\frac{dp_n}{dt} = \beta(n-1)p_{n-1} + \mu(n+1)p_{n+1} - (\beta + \mu)n p_n, \quad n \geq 1, \quad (2.10)$$

with the usual convention that terms with negative indices are omitted. The first line records that the only way to enter 0 is by a death from state 1; the second records births from $n - 1$, deaths from $n + 1$, and the total exit rate from n .

As it stands the master equation is an infinite linear system for the law of X_t , and two standard compressions of it will matter later. Taking moments against n produces ordinary

differential equations for means and, with more work, for higher moments. Passing instead to the probability generating function

$$G(z, t) = \sum_{n=0}^{\infty} p_n(t) z^n$$

converts the same system into a single first-order partial differential equation for G , which is the natural continuous-time counterpart of the functional iteration of discrete branching. We do not develop that partial differential equation in full here; it appears first for linear birth–death in [section 4.3](#), and then in greater generality under the method of characteristics in [section 5](#).

The same flux bookkeeping extends immediately to multi-type or multi-compartment counts. Writing $p_{i,j}(t) = \mathbb{P}(X_t = i, Y_t = j)$ for the two-compartment process of [section 5](#), each admissible transition contributes a gain term at the arrival state and a loss term at the departure state. Generating-function methods compress that countable system into a first-order PDE whose characteristic curves are ordinary differential equations in the auxiliary variables (x, y) . The preliminary vocabulary of this section is therefore already the language of the exact solutions derived later.

3 Galton–Watson survival and the limiting conditional law

3.1 Offspring PGFs and functional iteration

Let ξ be a non-negative integer-valued random variable with offspring probabilities

$$\mathbb{P}(\xi = j) = p_j, \quad f(s) = \mathbb{E}[s^\xi] = \sum_{j=0}^{\infty} p_j s^j, \quad 0 \leq s \leq 1. \quad (3.1)$$

The Galton–Watson process with $Z_0 = k$ is defined recursively by

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i}, \quad (3.2)$$

where the $\xi_{n,i}$ are independent copies of ξ and an empty sum is zero. Its mean offspring number is

$$m = f'(1) = \mathbb{E}[\xi].$$

This model is subcritical, critical, or supercritical according as $m < 1$, $m = 1$, or $m > 1$ [1].

Write $f_0(s) = s$ and $f_{n+1}(s) = f(f_n(s))$ for the functional iterates of f . The branching property gives the basic identity

$$\mathbb{E}[s^{Z_n} \mid Z_0 = k] = [f_n(s)]^k. \quad (3.3)$$

Indeed, one ancestor has PGF f_n , while the k ancestral lineages evolve independently. Differentiation at $s = 1$ then gives

$$\mathbb{E}[Z_n \mid Z_0 = k] = km^n. \quad (3.4)$$

For one ancestor, let

$$q_n = \mathbb{P}(Z_n = 0) = f_n(0), \quad S_n = \mathbb{P}(Z_n > 0) = 1 - q_n. \quad (3.5)$$

Thus $q_0 = 0$, $S_0 = 1$, and $q_{n+1} = f(q_n)$. Since extinction is absorbing, (q_n) is non-decreasing and has a limit q . The continuity of f shows that $f(q) = q$; more precisely, q is the smallest fixed point of f in $[0, 1]$ [1].

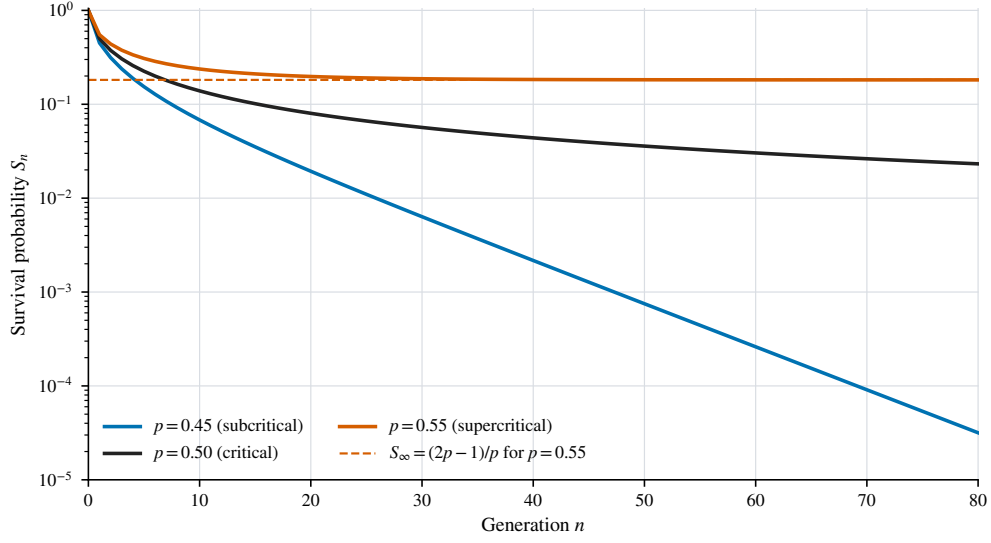


Figure 1: Exact survival probabilities obtained from [equation \(3.8\)](#). Subcritical survival decays geometrically at late generations, critical survival decays algebraically, and supercritical survival approaches the positive limit in [equation \(3.9\)](#).

3.2 The binary death-or-division process

The working model assigns either zero or two descendants to each individual:

$$p_0 = 1 - p, \quad p_2 = p, \quad f(s) = 1 - p + ps^2. \quad (3.6)$$

Its mean and variance are

$$m = 2p, \quad \sigma_\xi^2 = 4p(1 - p) = m(2 - m). \quad (3.7)$$

Substitution of $q_n = 1 - S_n$ into $q_{n+1} = f(q_n)$ yields the exact logistic recurrence

$$q_{n+1} = 1 - p + pq_n^2, \quad S_{n+1} = mS_n - pS_n^2 = mS_n \left(1 - \frac{S_n}{2}\right). \quad (3.8)$$

The fixed-point equation factors as

$$f(q) - q = (q - 1)\{pq - (1 - p)\}.$$

Consequently,

$$q = \begin{cases} 1, & 0 \leq p \leq \frac{1}{2}, \\ \frac{1-p}{p}, & \frac{1}{2} < p \leq 1, \end{cases} \quad \mathbb{P}(Z_n > 0 \text{ for all } n) = \begin{cases} 0, & p \leq \frac{1}{2}, \\ \frac{2p-1}{p}, & p > \frac{1}{2}. \end{cases} \quad (3.9)$$

The factor $1/p$ in the supercritical survival probability is essential. A positive expected growth rate does not itself give the establishment probability.

3.3 Critical survival and extinction time

At criticality, $p = \frac{1}{2}$ and $m = 1$, so

$$S_{n+1} = S_n \left(1 - \frac{S_n}{2}\right). \quad (3.10)$$

Ultimate extinction is certain, hence $S_n \downarrow 0$. The recurrence nevertheless decays slowly enough to give an infinite mean extinction time.

Proposition 3.1 (Critical survival asymptotic). *For the binary process with $p = \frac{1}{2}$ and $Z_0 = 1$,*

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty). \quad (3.11)$$

Proof. From equation (3.10),

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{2(1 - S_n/2)} \rightarrow \frac{1}{2}.$$

Since $1/S_n \rightarrow \infty$, the Stolz–Cesàro theorem gives

$$\lim_{n \rightarrow \infty} \frac{1/S_n}{n} = \lim_{n \rightarrow \infty} \left(\frac{1}{S_{n+1}} - \frac{1}{S_n} \right) = \frac{1}{2},$$

which is equivalent to equation (3.11). $\square$

Let

$$T = \inf\{n \geq 0 : Z_n = 0\}$$

be the extinction generation. The tail-sum identity gives

$$\mathbb{E}[T] = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} S_n. \quad (3.12)$$

By equation (3.11), the tail of this series is asymptotic to twice the harmonic series. Thus extinction occurs almost surely but $\mathbb{E}[T] = \infty$. Moreover,

$$\mathbb{P}(T = n + 1) = S_n - S_{n+1} = \frac{S_n^2}{2} \sim \frac{2}{n^2},$$

which makes explicit the heavy extinction-time tail.

3.4 Critical total progeny

The extinction generation records how long a lineage remains active; the total progeny

$$\mathcal{T} := \sum_{n=0}^{\infty} Z_n$$

records how many individuals it contains over its entire lifetime. Critical binary branching gives an exact power law, without a continuum approximation.

Proposition 3.2 (Critical total-progeny law). *For $p = \frac{1}{2}$ and $Z_0 = 1$,*

$$\mathbb{P}(\mathcal{T} = 2k + 1) = \frac{C_k}{2^{2k+1}}, \quad C_k = \frac{1}{k+1} \binom{2k}{k}, \quad k \geq 0, \quad (3.13)$$

and, along the odd support,

$$\mathbb{P}(\mathcal{T} = r) \sim \sqrt{\frac{2}{\pi}} r^{-3/2} \quad (r \rightarrow \infty, \text{ } r \text{ odd}). \quad (3.14)$$

Proof. A finite binary family tree with total size $2k + 1$ has k dividing individuals and $k + 1$ leaves. There are C_k rooted ordered full binary trees with this composition, and each has probability $2^{-(2k+1)}$ because every individual independently divides or dies with probability $1/2$. This proves equation (3.13). Stirling's formula gives $C_k \sim 4^k / (\sqrt{\pi} k^{3/2})$; substituting $r = 2k + 1$ yields equation (3.14). $\square$

The exponent in [equation \(3.14\)](#) is a lattice local limit: the even probabilities are exactly zero. A numerical histogram must therefore respect the odd support, and a family still alive at a finite simulation horizon is right-censored rather than observed to become extinct at that horizon.

3.5 Subcritical survival and the Kolmogorov constant

Now take $0 < p < \frac{1}{2}$, so $m = 2p \in (0, 1)$. The linearisation of [equation \(3.8\)](#) near $S = 0$ suggests S_n should be proportional to m^n . The proportionality constant exists exactly and has a convergent product representation.

Theorem 3.3 (Kolmogorov constant for binary offspring). *For $0 < p < \frac{1}{2}$,*

$$A(p) := \lim_{n \rightarrow \infty} \frac{S_n}{m^n} = \prod_{j=0}^{\infty} \left(1 - \frac{S_j}{2}\right) = \frac{1}{2} \prod_{j=1}^{\infty} \left(1 - \frac{S_j}{2}\right) \in (0, \frac{1}{2}). \quad (3.15)$$

Consequently,

$$S_n \sim A(p)m^n. \quad (3.16)$$

Proof. Set $a_n = S_n/m^n$. Equation (3.8) gives

$$a_{n+1} = a_n \left(1 - \frac{S_n}{2}\right),$$

and iteration from $a_0 = S_0 = 1$ gives

$$a_n = \prod_{j=0}^{n-1} \left(1 - \frac{S_j}{2}\right).$$

Every factor is positive. In addition, $S_{n+1} \leq mS_n$, so $S_n \leq m^n$ and $\sum_n S_n < \infty$. The standard criterion for infinite products therefore shows that the product converges to a strictly positive limit. Finally, its first factor is $1 - S_0/2 = 1/2$. $\square$

Corollary 3.4 (Endpoint behaviour). *The Kolmogorov constant has the one-sided limits*

$$\lim_{p \downarrow 0} A(p) = \frac{1}{2}, \quad \lim_{p \uparrow 1/2} A(p) = 0. \quad (3.17)$$

Proof. For $j \geq 1$, [equation \(3.8\)](#) gives $S_j \leq p(2p)^{j-1}$, and hence

$$\sum_{j=1}^{\infty} S_j \leq \frac{p}{1-2p}.$$

The product inequality $0 \leq 1 - \prod_j (1 - x_j) \leq \sum_j x_j$, applied with $x_j = S_j/2$, now shows that the product after the first factor in [equation \(3.15\)](#) tends to one as $p \downarrow 0$.

For the other endpoint, fix N and let

$$A_N(p) = \prod_{j=0}^N \left(1 - \frac{S_j(p)}{2}\right).$$

Then $0 < A(p) \leq A_N(p)$, while continuity of the finite recurrence gives $A_N(p) \rightarrow A_N(1/2)$ as $p \uparrow 1/2$. At criticality the product recurrence yields $A_N(1/2) = S_{N+1}(1/2)$, which tends to zero by [equation \(3.11\)](#). Letting $N \rightarrow \infty$ proves the second limit. $\square$

The constant $A(p)$ is the classical *Kolmogorov constant* for this subcritical Galton–Watson process [4]. Its defining survival ratio applies on $0 < p < \frac{1}{2}$; at $p = 0$ that ratio is degenerate, while [equation \(3.17\)](#) supplies the continuous endpoint values and shows that the conditional population scale diverges as $p \uparrow \frac{1}{2}$.

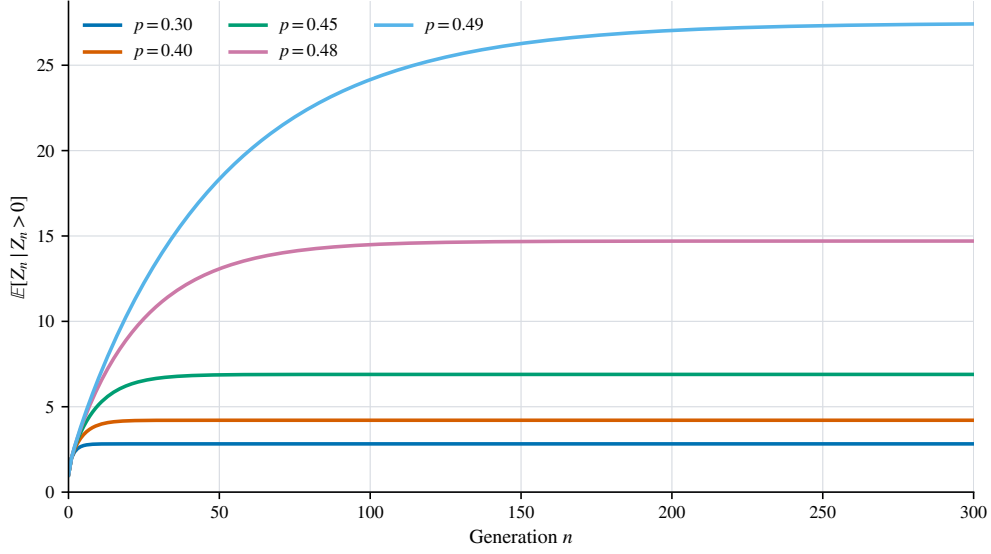


Figure 2: Finite-generation conditional mean m^n/S_n for representative subcritical division probabilities. Each curve increases to the Yaglom mean $1/A(p)$. Convergence becomes slow near $p = 1/2$, where the conditional scale is large.

3.6 Yaglom limit and conditional moments

For $0 < p < \frac{1}{2}$, conditioning on the rare event $\{Z_n > 0\}$ produces a non-degenerate limiting conditional law. Its finite-generation PGF is

$$B_n(s) = \mathbb{E}[s^{Z_n} \mid Z_n > 0] = \frac{f_n(s) - f_n(0)}{1 - f_n(0)}. \quad (3.18)$$

The Yaglom limit $B(s) = \lim_{n \rightarrow \infty} B_n(s)$ exists for the present finite-support offspring law and is the unique finite-mean PGF satisfying

$$1 - B(f(s)) = m\{1 - B(s)\}, \quad B(0) = 0, \quad B(1) = 1. \quad (3.19)$$

This limiting law is independent of the finite, positive initial cohort size [1, 4].

The conditional mean follows without first finding B . Since $Z_n = 0$ on extinction,

$$\mathbb{E}[Z_n \mid Z_n > 0] = \frac{\mathbb{E}[Z_n]}{S_n} = \frac{m^n}{S_n} \longrightarrow \frac{1}{A(p)}. \quad (3.20)$$

Thus the inverse Kolmogorov constant is the mean of the limiting conditional law.

The second moment supplies a useful consistency check. The law-of-total-variance recursion gives

$$\text{Var}(Z_{n+1}) = m^2 \text{Var}(Z_n) + \sigma_\xi^2 \mathbb{E}[Z_n],$$

and $\text{Var}(Z_0) = 0$, one obtains

$$\text{Var}(Z_n) = \sigma_\xi^2 m^{n-1} \frac{1 - m^n}{1 - m} = (2 - m)m^n \frac{1 - m^n}{1 - m}. \quad (3.21)$$

It follows that

$$\lim_{n \rightarrow \infty} \mathbb{E}[Z_n^2 \mid Z_n > 0] = \frac{2 - m}{A(p)(1 - m)}, \quad (3.22)$$

$$\lim_{n \rightarrow \infty} \text{Var}(Z_n \mid Z_n > 0) = \frac{2 - m}{A(p)(1 - m)} - \frac{1}{A(p)^2}. \quad (3.23)$$

The same result follows by differentiating [equation \(3.19\)](#). In particular, $B''(1) = \{A(p)(1 - m)\}^{-1}$.

3.7 Koenigs linearisation and certified computation

The survival recurrence becomes a standard logistic map after the rescaling

$$w_n = \frac{S_n}{2}, \quad w_{n+1} = L_m(w_n) := mw_n(1 - w_n), \quad w_0 = \frac{1}{2}.$$

For $0 < m < 1$, the origin is attracting. Its normalised Koenigs function ψ_m satisfies

$$\psi_m(L_m(w)) = m\psi_m(w), \quad \psi_m(0) = 0, \quad \psi'_m(0) = 1, \quad (3.24)$$

and, by Koenigs' linearisation theorem [3], may be represented on this real basin by

$$\psi_m(w) = \lim_{n \rightarrow \infty} \frac{L_m^{\circ n}(w)}{m^n}. \quad (3.25)$$

Since $L_m^{\circ n}(1/2) = w_n$, comparison with [equation \(3.15\)](#) gives the exact identity

$$A(p) = 2\psi_{2p}\left(\frac{1}{2}\right). \quad (3.26)$$

This identity explains why direct functional iteration and the infinite product compute the same quantity.

No general explicit expression for the Kolmogorov constant is currently known [4]. For each fixed $0 < m < 1$, the non-linear logistic Koenigs function is differentially transcendental as a function of its complex state variable [2]. This fixed-map result does not, by itself, prove that the scalar parameter map $p \mapsto A(p)$ has no elementary expression. Equation (3.26) should therefore be read as an exact representation, not as an impossibility theorem.

For numerical work, define

$$M_n = \frac{m^n}{S_n}.$$

The sequence (M_n) increases to $M_\infty = 1/A(p)$. Specialising a rigorous branching-process bound to $f(s) = 1 - p + ps^2$ gives, for $n \geq 1$,

$$0 < M_\infty - M_n \leq \frac{m^n}{2(1 - m)q_n}, \quad q_n = 1 - S_n. \quad (3.27)$$

Thus recurrence iteration can be stopped against a requested absolute error rather than an arbitrary population threshold [4]. This control matters near criticality, where a fixed iteration cap or a rule based only on S_n can substantially overestimate $A(p)$.

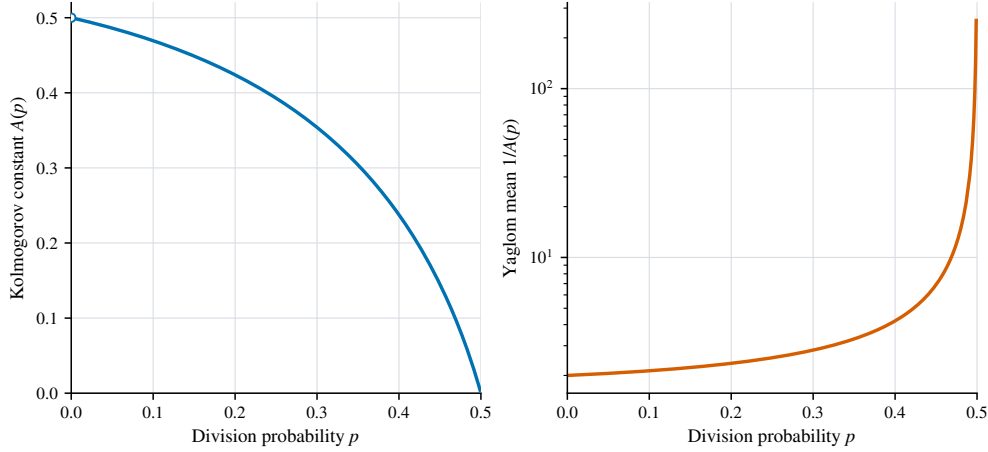


Figure 3: The Kolmogorov constant $A(p)$, computed from the exact product in [equation \(3.15\)](#) with tail control, and its reciprocal Yaglom mean. The endpoint at $p = 0$ is the continuous extension $A(0+) = 1/2$; the defining survival ratio is used only for $0 < p < 1/2$.

4 Initial cohorts, catastrophe exposure, and continuous-time comparison

4.1 Survival from a finite cohort

Suppose the binary Galton–Watson process begins with $Z_0 = k \geq 1$. Extinction by generation n requires all k ancestral lineages to be extinct, and those lineages are independent. Therefore

$$\mathbb{P}(Z_n = 0 \mid Z_0 = k) = q_n^k, \quad S_n^{(k)} := \mathbb{P}(Z_n > 0 \mid Z_0 = k) = 1 - q_n^k = 1 - (1 - S_n)^k. \quad (4.1)$$

This exact expression separates two effects that are easily conflated: the criticality parameter p determines the fate of each lineage, while the initial count k determines how many independent opportunities for persistence are present.

In the subcritical case, $S_n \rightarrow 0$ and a binomial expansion gives

$$S_n^{(k)} = kS_n + O(S_n^2) \sim kA(p)m^n. \quad (4.2)$$

Since $\mathbb{E}[Z_n \mid Z_0 = k] = km^n$, the factors of k cancel after conditioning:

$$\mathbb{E}[Z_n \mid Z_n > 0, Z_0 = k] = \frac{km^n}{S_n^{(k)}} \longrightarrow \frac{1}{A(p)}. \quad (4.3)$$

Thus a larger inoculum makes late survival more likely but does not alter the limiting conditional law reached by a surviving finite cohort.

In the supercritical case, [equation \(3.9\)](#) instead gives the ultimate establishment probability

$$\mathbb{P}(Z_n > 0 \text{ for all } n \mid Z_0 = k) = 1 - \left(\frac{1-p}{p}\right)^k, \quad p > \frac{1}{2}. \quad (4.4)$$

Close to criticality, even the one-lineage probability is small, but repeated independent opportunities can change the cohort-level probability substantially. This is the precise sense in which early cohort size can dominate establishment in a slightly supercritical process.

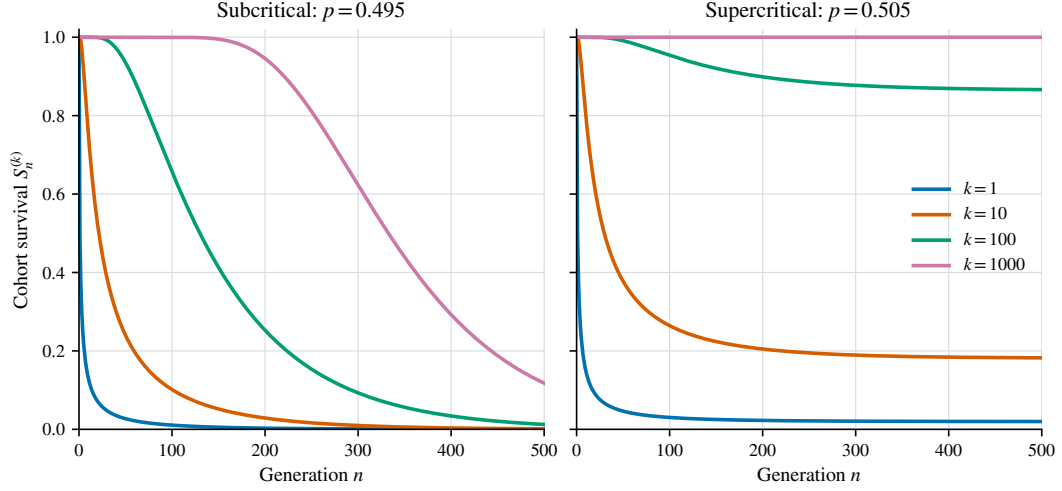


Figure 4: Survival probability $S_n^{(k)}$ for initial cohort sizes $k \in \{1, 10, 100, 1000\}$. The subcritical process ($p = 0.495$) eventually becomes extinct for every finite k , whereas the slightly supercritical process ($p = 0.505$) approaches the positive establishment probability in equation (4.4). All curves are evaluated from the exact recurrence, without Monte Carlo error.

4.2 A state-dependent catastrophe is not a mean-field hazard

A global catastrophe initiated by any individual provides a useful warning about state-dependent hazards. Once a trigger occurs, every surviving lineage is stopped together, so the realised killed lineages are coupled and the event probability must be defined with care. Let $\kappa \in [0, 1]$ be the probability that an individual triggers catastrophe at a generational check. Checks occur before reproduction, independently between individuals and generations; if no individual triggers catastrophe, offspring are then generated with PGF f .

Let a_n be the probability that no catastrophe occurs during the first n checks, starting from one ancestor and regardless of whether the population becomes naturally extinct. Conditioning on the first check and on the ancestor's offspring gives

$$a_0 = 1, \quad a_{n+1} = (1 - \kappa)f(a_n). \quad (4.5)$$

Although the killed populations share a global stopping event, their potential family trees and trigger variables factor before any trigger occurs. Thus, for k initial ancestors, the probability of no catastrophe is a_n^k . In the special deterministic process in which every individual always divides, $Z_j = 2^j$ before catastrophe and

$$a_n = (1 - \kappa)^{1+2+\dots+2^{n-1}} = (1 - \kappa)^{2^n - 1}. \quad (4.6)$$

By contrast, $1 - (1 - \kappa)^{2^{n-1}}$ is only the conditional probability of a catastrophe at the n th check given that all earlier checks were passed.

When demographic death makes Z_j random, define the cumulative exposure

$$C_n = \sum_{j=0}^{n-1} Z_j$$

for the process without catastrophe. Then

$$a_n = \mathbb{E}[(1 - \kappa)^{C_n}], \quad \mathbb{P}(\text{no catastrophe and } Z_n > 0) = \mathbb{E}[(1 - \kappa)^{C_n} \mathbf{1}_{\{Z_n > 0\}}]. \quad (4.7)$$

The nonlinear expectation in [equation \(4.7\)](#) is not equal to $(1 - \kappa)^{\mathbb{E}[C_n]}$. Replacing the random exposure by its mean therefore does not close the model; a marked or multivariate PGF is required. This observation motivates the compartmental PGFs in [section 5](#).

4.3 The continuous-time linear birth–death process

For comparison, let N_t be a continuous-time population in which each individual gives birth at rate $\lambda \geq 0$ and dies at rate $\mu \geq 0$. From state z , the transitions $z \rightarrow z + 1$ and $z \rightarrow z - 1$ therefore occur at rates λz and μz . Starting from one individual, the extinction probability by time t is

$$q(t) = \begin{cases} \frac{\mu\{1 - e^{-(\mu-\lambda)t}\}}{\mu - \lambda e^{-(\mu-\lambda)t}}, & \lambda \neq \mu, \\ \frac{\lambda t}{1 + \lambda t}, & \lambda = \mu. \end{cases} \quad (4.8)$$

For $N_0 = k$, the branching property again gives $\mathbb{P}(N_t = 0 \mid N_0 = k) = q(t)^k$. The mean is

$$\mathbb{E}[N_t \mid N_0 = k] = k e^{(\lambda - \mu)t}. \quad (4.9)$$

The subcritical regime is $\lambda < \mu$. Setting $d = \mu - \lambda > 0$, the one-ancestor survival probability and conditional mean are

$$S_{\text{ct}}(t) = \frac{de^{-dt}}{\mu - \lambda e^{-dt}}, \quad (4.10)$$

$$\mathbb{E}[N_t \mid N_t > 0, N_0 = 1] = \frac{e^{-dt}}{S_{\text{ct}}(t)} = \frac{\mu - \lambda e^{-dt}}{d} \rightarrow \frac{\mu}{\mu - \lambda}. \quad (4.11)$$

The limiting conditional law is shifted geometric:

$$\mathbb{P}(N_\infty^* = j) = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^{j-1}, \quad j \geq 1, \quad (4.12)$$

whose mean agrees with [equation \(4.11\)](#). These standard Markov branching results may also be recovered from the PGF equation used in [section 5.4 \[1\]](#).

It is useful to normalise [equation \(4.10\)](#) by its exponential factor:

$$A_{\text{ct}} := \lim_{t \rightarrow \infty} \frac{S_{\text{ct}}(t)}{e^{-(\mu-\lambda)t}} = \frac{\mu - \lambda}{\mu}. \quad (4.13)$$

When $\lambda + \mu > 0$, let $\rho = \lambda/(\lambda + \mu)$ denote the dimensionless birth fraction. Then

$$A_{\text{ct}}(\rho) = \frac{1 - 2\rho}{1 - \rho}, \quad 0 \leq \rho < \frac{1}{2}. \quad (4.14)$$

At criticality, $\lambda = \mu > 0$, [equation \(4.8\)](#) gives

$$\mathbb{P}(N_t > 0) = \frac{1}{1 + \lambda t} \sim \frac{1}{\lambda t}.$$

The continuous-time critical process therefore mirrors the discrete process: extinction is certain, the conditional population scale grows linearly, and the mean extinction time diverges.

Table 1: Events in the absorption–birth–death model. Terms of order $o(\Delta t)$ are omitted from the corresponding infinitesimal transition probabilities.

Event	State change from (i, j)	Rate
Absorption	$(i, j) \rightarrow (i + 1, j - 1)$	αj
Intracellular birth	$(i, j) \rightarrow (i + 1, j)$	βi
Intracellular death	$(i, j) \rightarrow (i - 1, j)$	μi

4.4 Why the discrete and continuous amplitudes differ

The symbols $A(p)$ and $A_{\text{ct}}(\rho)$ describe analogous asymptotic normalisations, but they belong to different stochastic clocks and event rules. In the discrete binary process, an individual present at generation n must divide to be represented at generation $n + 1$; conditional on one-step survival, the new count is two. Accordingly, the continuous extension of the discrete constant is $A(0+) = 1/2$, and the limiting conditional mean approaches 2 as $p \downarrow 0$.

In continuous time with $\lambda = 0$, by contrast, a surviving individual is simply the original individual whose death clock has not yet rung. Its conditional count is one, so $A_{\text{ct}}(0) = 1$. There is therefore no reason for the two amplitude curves to coincide or to be related by a constant factor away from criticality. What is shared is the structural statement: survival probability equals a decaying mean scale multiplied by an amplitude, and the inverse amplitude is the limiting mean conditional on survival.

5 Multivariate PGFs and the method of characteristics

The branching calculations above concern a single population count. We now use the same PGF structure to follow a pathogenic cohort as it moves from an extracellular compartment into a host cell. The models form a hierarchy: absorption is introduced first, intracellular death is then added, and intracellular replication is added last. This order keeps the probabilistic meaning of each term visible.

5.1 A two-compartment Markov model

Let X_t and Y_t denote the intracellular and extracellular particle counts. For a fixed $N \in \mathbb{N}$, we take

$$(X_0, Y_0) = (0, N)$$

and impose three modelling assumptions: particles act independently, the host has no capacity limit, and intracellular descendants do not return to the extracellular compartment. The non-negative rates α , β , and μ govern absorption, intracellular birth, and intracellular death, respectively, as listed in [table 1](#).

For a test function h on the state space, the generator is

$$\begin{aligned} \mathcal{L}h(i, j) = & \alpha j \{h(i + 1, j - 1) - h(i, j)\} \\ & + \beta i \{h(i + 1, j) - h(i, j)\} + \mu i \{h(i - 1, j) - h(i, j)\}. \end{aligned} \quad (5.1)$$

Define the bivariate PGF

$$G(x, y, t) = \mathbb{E}[x^{X_t} y^{Y_t}] = \sum_{i=0}^{\infty} \sum_{j=0}^N p_{i,j}(t) x^i y^j, \quad p_{i,j}(t) = \mathbb{P}(X_t = i, Y_t = j). \quad (5.2)$$

for $(x, y) \in [0, 1]^2$. Applying [equation \(5.1\)](#) to $h(i, j) = x^i y^j$ and using $\frac{d}{dt} \mathbb{E}[h(X_t, Y_t)] = \mathbb{E}[\mathcal{L}h(X_t, Y_t)]$ gives

$$\frac{\partial G}{\partial t} = a(x) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad a(x) = \beta x^2 - (\beta + \mu)x + \mu, \quad (5.3)$$

with initial condition

$$G(x, y, 0) = y^N. \quad (5.4)$$

At $(x, y) = (1, 1)$ both transport coefficients in [equation \(5.3\)](#) vanish; together with the initial value, this gives the normalisation $G(1, 1, t) = 1$.

Written as

$$G_t - a(x)G_x + \alpha(y - x)G_y = 0,$$

[equation \(5.3\)](#) has characteristic equations

$$\frac{dt}{ds} = 1, \quad \frac{dx}{ds} = -a(x), \quad \frac{dy}{ds} = \alpha(y - x), \quad \frac{dG}{ds} = 0. \quad (5.5)$$

Thus G is constant along a characteristic, and the solution at (x, y, t) is obtained by tracing that characteristic back to the initial surface $t = 0$. The following special cases make this construction explicit.

5.2 Absorption without intracellular dynamics

Set $\beta = \mu = 0$. Equation [\(5.3\)](#) reduces to

$$G_t = \alpha(x - y)G_y. \quad (5.6)$$

Here x is constant along a characteristic and y solves

$$\frac{dy}{ds} = \alpha(y - x).$$

Tracing the terminal point (x, y) at time t back to time zero gives

$$y_0 = x + (y - x)e^{-\alpha t}.$$

Since the initial PGF is y_0^N ,

$$G_{\text{abs}}(x, y, t) = \left[\{1 - e^{-\alpha t}\}x + e^{-\alpha t}y \right]^N. \quad (5.7)$$

The same formula follows directly from independence. Each initial particle remains extracellular with probability $e^{-\alpha t}$ and has been absorbed with probability $1 - e^{-\alpha t}$. Consequently,

$$\mathbb{P}(X_t = i, Y_t = j) = \mathbf{1}_{\{i+j=N\}} \frac{N!}{i!j!} \{1 - e^{-\alpha t}\}^i e^{-\alpha t j}. \quad (5.8)$$

In particular,

$$\mathbb{E}[Y_t] = N e^{-\alpha t}, \quad \mathbb{E}[X_t] = N \{1 - e^{-\alpha t}\}.$$

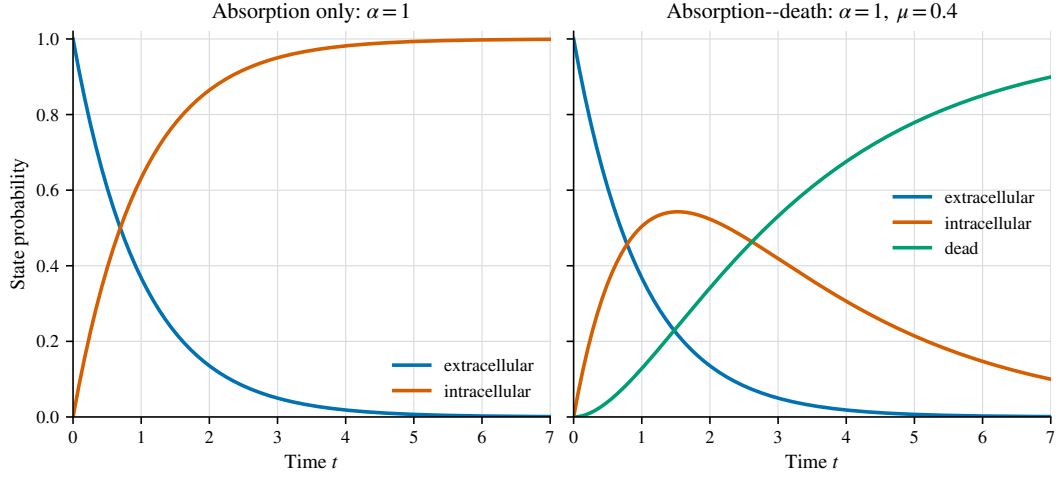


Figure 5: One-particle state probabilities. Left: absorption only, with $\alpha = 1$. Right: absorption followed by intracellular death, with $\alpha = 1$ and $\mu = 0.4$. The death state accumulates the probability that has left both living compartments.

5.3 Absorption followed by intracellular death

Now set $\beta = 0$ and retain $\mu \geq 0$. One initially extracellular particle can occupy only three states at time t : extracellular, intracellular, or dead. Denote the corresponding probabilities by $\pi_E(t)$, $\pi_I(t)$, and $\pi_D(t)$. The first is

$$\pi_E(t) = e^{-\alpha t}.$$

Conditioning on an absorption time s gives

$$\pi_I(t) = \int_0^t \alpha e^{-\alpha s} e^{-\mu(t-s)} ds = \begin{cases} \frac{\alpha}{\mu - \alpha} \{e^{-\alpha t} - e^{-\mu t}\}, & \alpha \neq \mu, \\ \alpha t e^{-\alpha t}, & \alpha = \mu, \end{cases} \quad (5.9)$$

and $\pi_D(t) = 1 - \pi_I(t) - \pi_E(t)$. Independence of the N initial particles therefore yields

$$G_{\text{ad}}(x, y, t) = \{\pi_D(t) + \pi_I(t)x + \pi_E(t)y\}^N. \quad (5.10)$$

Substitution verifies that [equation \(5.10\)](#) solves

$$G_t = \mu(1 - x)G_x + \alpha(x - y)G_y$$

with initial condition y^N . The characteristic method and the direct probabilistic construction are therefore two descriptions of the same solution, rather than competing approximations.

The complete state probabilities are multinomial:

$$p_{i,j}(t) = \frac{N!}{i!j!(N-i-j)!} \pi_I(t)^i \pi_E(t)^j \pi_D(t)^{N-i-j}, \quad i, j \geq 0, \quad i + j \leq N. \quad (5.11)$$

They are zero when $i + j > N$.

5.4 Absorption followed by intracellular birth and death

When $\beta > 0$, a single absorbed particle can generate an unbounded number of intracellular descendants. This removes the finite three-state shortcut, but it does not remove the branching property. The cleanest exact solution conditions on the absorption time.

Let

$$F(x, t) = \mathbb{E}[x^{W_t} \mid W_0 = 1] \quad (5.12)$$

be the PGF of a linear birth–death process with per-particle birth rate β and death rate μ . A first-event argument gives the Riccati equation

$$F_t = \beta F^2 - (\beta + \mu)F + \mu, \quad F(x, 0) = x. \quad (5.13)$$

For $\beta \neq \mu$, its solution can be written without introducing singular root labels as

$$F(x, t) = \frac{e^{(\mu-\beta)t}(\beta x - \mu) + \mu(1-x)}{e^{(\mu-\beta)t}(\beta x - \mu) + \beta(1-x)}. \quad (5.14)$$

At $\beta = \mu = \gamma$, the continuous limit is

$$F(x, t) = \frac{x + \gamma t(1-x)}{1 + \gamma t(1-x)}. \quad (5.15)$$

Both formulae satisfy $F(1, t) = 1$ and $F(x, 0) = x$.

For one particle initially outside the cell, non-absorption by time t contributes $e^{-\alpha t}y$ to its bivariate PGF. If absorption occurs at time $s \in [0, t]$, with density $\alpha e^{-\alpha s}$, the intracellular descendants evolve for the remaining time $t - s$ and contribute $F(x, t - s)$. Hence

$$g(x, y, t) = e^{-\alpha t}y + \int_0^t \alpha e^{-\alpha s} F(x, t - s) ds. \quad (5.16)$$

The N initial particles generate independent descendant families, so the full solution is simply

$$G_{\text{abd}}(x, y, t) = g(x, y, t)^N. \quad (5.17)$$

Proposition 5.1. *The function in equation (5.17) is the normalised solution of equations (5.3) and (5.4).*

Proof. The branching semigroup for F also satisfies

$$F_t = a(x)F_x, \quad a(x) = \beta x^2 - (\beta + \mu)x + \mu.$$

Differentiating equation (5.16), using $F(x, 0) = x$, gives

$$g_t = \alpha e^{-\alpha t}(x - y) + a(x)g_x = a(x)g_x + \alpha(x - y)g_y,$$

because $g_y = e^{-\alpha t}$. The chain rule then shows that $G = g^N$ satisfies the same linear PDE. At $t = 0$, $g = y$ and hence $G = y^N$; at $(x, y) = (1, 1)$, the two mutually exclusive absorption-time cases in equation (5.16) sum to one. $\square$

The integral in equation (5.16) can be reduced further for generic parameter choices, but the one-dimensional integral is the more stable exact representation. It remains valid at equal rates, preserves normalisation transparently, and avoids branch choices and removable singularities.

5.5 Coefficient extraction and moment checks

For any analytic bivariate PGF, the state probabilities are

$$p_{i,j}(t) = \frac{1}{i!j!} \left. \frac{\partial^{i+j} G}{\partial x^i \partial y^j} \right|_{x=y=0}. \quad (5.18)$$

If $G = (c_0 + c_x x + c_y y)^N$, the multinomial theorem gives the more explicit formula

$$[x^i y^j]G = \frac{N!}{i!j!(N-i-j)!} c_x^i c_y^j c_0^{N-i-j}, \quad i+j \leq N. \quad (5.19)$$

The falling factorial in [equation \(5.19\)](#) is essential; replacing it by N^{i+j} does not differentiate a power correctly. [Equations \(5.8\)](#) and [\(5.11\)](#) are immediate special cases.

For the absorption–birth–death model, write

$$b(x, t) = \int_0^t \alpha e^{-\alpha s} F(x, t-s) ds, \quad g(x, y, t) = b(x, t) + e^{-\alpha t} y.$$

Since the extracellular count cannot exceed N ,

$$p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} [x^i] b(x, t)^{N-j}, \quad 0 \leq j \leq N. \quad (5.20)$$

This reduces coefficient recovery to a univariate series problem even though the intracellular count is unbounded.

Two mean identities provide inexpensive checks on analytical or numerical implementations. Differentiating the PGF at $(1, 1)$, or taking expectations in the transition rates, gives

$$\mathbb{E}[Y_t] = N e^{-\alpha t} \quad (5.21)$$

and

$$\mathbb{E}[X_t] = \begin{cases} \frac{\alpha N}{\alpha + \beta - \mu} \left\{ e^{(\beta - \mu)t} - e^{-\alpha t} \right\}, & \alpha + \beta - \mu \neq 0, \\ \alpha N t e^{-\alpha t}, & \alpha + \beta - \mu = 0. \end{cases} \quad (5.22)$$

Together with $G(1, 1, t) = 1$ and $G(x, y, 0) = y^N$, these identities test normalisation, signs, equal-rate limits, and the placement of every exponential.

5.6 Synthesis

The discrete and continuous models are connected by more than notation. In both, independence converts an initial cohort into a power of a one-ancestor PGF. Extinction and conditioning are controlled by evaluations of that PGF at zero, while moments are its derivatives at one. Nonlinearity enters differently: functional iteration produces the Kolmogorov constant in discrete time, whereas the continuous-time birth–death semigroup solves a Riccati equation. Once absorption couples two counts, the generator converts those same mechanisms into a first-order PDE, and characteristics recover the probabilistic decomposition.

This common structure also fixes the appropriate level of approximation. A mean trajectory is adequate for the identities in [equations \(5.21\)](#) and [\(5.22\)](#), but it cannot replace the path distribution in a survival or catastrophe probability. Conversely, an exact PGF need not be expanded into a cumbersome special-function expression when a normalised integral representation already exposes the model, its limiting cases, and a stable route to computation.

References

- [1] Krishna B. Athreya and Peter E. Ney. *Branching Processes*, volume 196 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin and Heidelberg, 1972. doi:[10.1007/978-3-642-65371-1](https://doi.org/10.1007/978-3-642-65371-1).
- [2] Paul-Georg Becker and Walter Bergweiler. Hypertranscendence of conjugacies in complex dynamics. *Mathematische Annalen*, 301(3):463–468, 1995. doi:[10.1007/BF01446639](https://doi.org/10.1007/BF01446639).
- [3] Gabriel Koenigs. Recherches sur les intégrales de certaines équations fonctionnelles. *Annales Scientifiques de l'École Normale Supérieure (3)*, 1:3–41, 1884. doi:[10.24033/asens.247](https://doi.org/10.24033/asens.247).
- [4] Anthony G. Pakes. Remarks on Kolmogorov's constant for simple branching processes. *Journal of Applied Probability*, pages 1–13, 2026. First View. doi:[10.1017/jpr.2026.10094](https://doi.org/10.1017/jpr.2026.10094).